

UNIT 4

Evidence , Collection and Data Seizure

Content

Unit 4	Evidence Collection and Data Seizure	06 Hours
Why Collect Evidence? Collection Options ,Obstacles, Types of Evidence — The Rules of Evidence, Volatile Evidence, General Procedure, Collection and Archiving, Methods of Collection, Artifacts, Collection Steps, Controlling Contamination: The Chain of Custody Duplication and Preservation of Digital Evidence: Preserving the Digital Crime Scene — Computer Evidence Processing Steps, Legal Aspects of Collecting and Preserving Computer Forensic Evidence Computer Image Verification and Authentication: Special Needs of Evidential Authentication, Practical Consideration, Practical Implementation.		

Why Collect Evidence ?

Definition of Evidence : -

- Information that proves something or helps prove something relevant to the case

Motivations for Collecting Evidence

1. Prevention in the Future

- Understanding what happened helps prevent similar incidents.
- Identifies weaknesses that allowed the attack.

2. Establishing Responsibility

- Evidence is required to hold the attacker accountable.
- Helps in legal or disciplinary actions.
- Supports the victim's social responsibility to report and prevent future harm.

3. Improving Security

- Information gathered after an attack can be analyzed.
- Helps strengthen systems against future attacks.

Collection Options

1. Remove the System from the Network

- Disconnect the compromised system and start collecting evidence.
- Risk: You may not have enough evidence available.
- Risk: The attacker may have installed a dead-man switch that erases data when the system goes offline.

2. Leave the System Online and Monitor

- Keep the system running and observe the attacker's activities.
- Risk: Monitoring may alert the intruder.
- The attacker may start covering their tracks, deleting or altering evidence

Obstacles in Evidence Collection

1. Admissibility Issues

- Computer transactions are fast, can occur from anywhere, and may be encrypted or anonymous.
- Lack of natural identifiers such as handwriting or signatures.
- Evidence trails can be altered, deleted, or may exist only temporarily.

2. Completeness Challenges

- Digital records can be easily modified or removed.
- Some logs or audit trails may be overwritten automatically by the system.
- Difficult to ensure the entire sequence of events is captured.

3. Believability Concerns

- Digital data can be manipulated, raising questions about its reliability.
- It is harder to prove that a particular individual performed a digital action.

4. Authenticity Problems

- Data may not be inherently trustworthy due to ease of alteration.
- Linking the evidence to the suspect becomes complex.

Types of evidence

1. Real Evidence

- Physical objects presented in court.
- Can be shown directly to the jury.
- Most powerful form of evidence.
- “Speaks for itself” without explanation.

2. Documentary Evidence

- Written or digital records (e.g., server logs, emails, database files).
- Must be authenticated because digital documents can be faked.
- Always present the original document, not a copy

3. Testimonial Evidence

- Statements given by a witness under oath.
- Provided in court or through depositions

- Helps support or validate other types of evidence.

4. Demonstrative Evidence

- Used to explain, illustrate, or recreate other evidence.
- Examples: charts, diagrams, models, reconstructions.
- Useful for explaining technical concepts to non-technical audiences.
- Does not “speak for itself”; it supports understanding.

Rules of evidence

1. Admissible

- Evidence must be acceptable in court.
- If evidence is not admissible, the case may fail.
- Poor evidence handling may allow a guilty person to go free.

2. Authentic

- Evidence must be genuine and directly related to the incident.
- Must prove that the evidence is connected to the case.
- Avoid collecting irrelevant but interesting information.

3. Complete

- Evidence should cover the entire incident, not just one side.
- Investigators must not assume guilt or innocence.
- Must rule out other suspects and explanations.
- Use proper forensic methods to reach clear conclusions.

4. Reliable

- Evidence must be trustworthy and consistent.
- Forensic tools and methods must be validated.
- Expert witness qualifications strengthen reliability.
- Collection and analysis should not raise doubts about authenticity.

5. Believable

- Evidence must be clear and understandable, even for non-technical jurors.
- Conclusions should be simple and logical.
- Other investigators should be able to use the same methods and reach the same

Volatile evidence

What is Volatile Evidence?

- Data that changes frequently and is lost when power is removed.
- Exists in active physical memory.
- Found in:
 - Physical RAM
 - CPU registers
 - Cache memory
 - Virtual memory
 - Temporary files
 - Peripheral device memory

Order of Volatility (Collect Most Volatile First)

1. Registers & Cache
2. Routing Tables
3. ARP Cache
4. Process Table

- Kernel Statistics & Modules
- Main Memory (RAM)
- Temporary File Systems
- Secondary Memory (HDD/SSD)
- Router Configuration
- Network Topology

General Procedure

1. Identification of Evidence : -

- Distinguish real evidence from false or irrelevant data.

2. Preservation of Evidence : -

- Keep evidence in its original condition as much as possible.
- Avoid contamination or alteration.

3. Evidence Analysis

- Requires deep knowledge of what to look for and how to extract it.
- Analyze patterns, logs, memory data, events, etc.

4. Evidence Presentation

- Present evidence clearly and simply.
- Must be understandable to non-technical people, including jurors.

Methods of Collection

Two Fundamental Methods:

1. Honeypotting

- Involves planting deceptive information to study attacker behavior.
- Helps identify the attacker's motives and methods.
- A replica system (honeypot) is created to lure the attacker for monitoring.

2. Freezing the Scene

- Capture a snapshot/image of the system at the time of compromise.
- Collect all crucial data onto non-volatile removable media.
- Generate cryptographic message digests (hashes) for each data item.
- Cross-check digests with originals to ensure accuracy and integrity.

Artifacts

What Are Artifacts?

- Items left behind by the attacker such as:
 - Malicious code
 - Malware
 - Active processes
 - Sniffer logs
 - Configuration changes
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Important Rule

- Never analyze evidence on the compromised machine itself.
- Artifacts can behave unpredictably; analysis should be done in a controlled environment.

Collection Steps

1. Find the Evidence

- Use a checklist to ensure nothing is missed.

2. Identify Relevant Data

- Determine which data is important to the investigation.

- Establish an Order of Volatility
- Collect the most volatile evidence first to avoid data loss.

4. Eliminate External Sources of Change

- Avoid modifying or altering the original data.

5. Gather Evidence

- Use proper forensic tools and methods for data collection.

6. Document Everything

- Maintain full documentation:
- Timestamps
- Digital signatures
- Signed statements
- This protects the credibility of your collection methods in court.

Controlling Contamination: The Chain of Custody

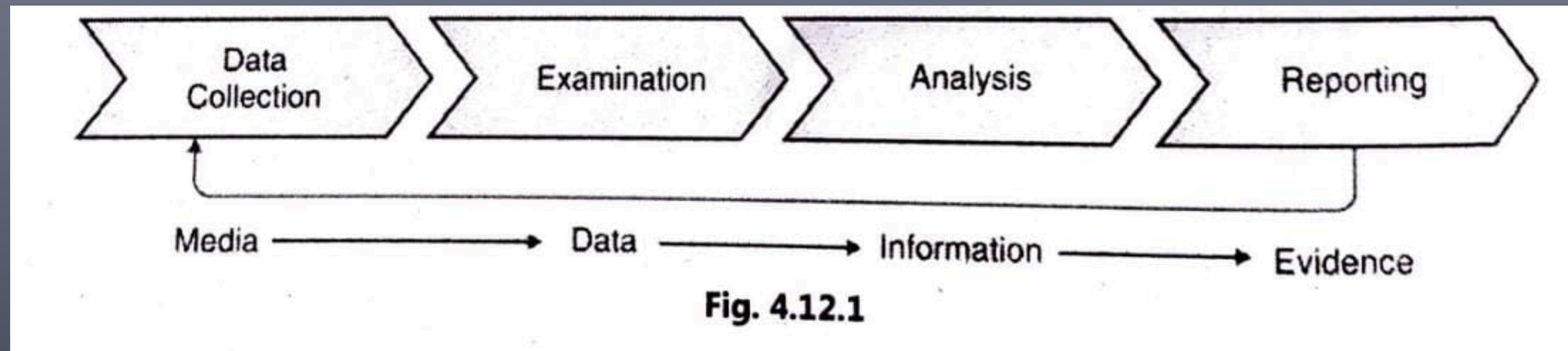
Chain of Custody – Meaning

- Documentation that tracks control, handling, custody, and ownership of evidence.
- Ensures evidence is stored in a tamper-proof manner.
- Maintains a complete record from the moment evidence is collected until it is presented in court.

Importance of Chain of Custody

- Shows where the evidence was at every stage.
- Prevents unauthorized access and contamination.
- Law enforcement uses secure property rooms to store evidence.
- Evidence must be checked-out and checked-in whenever reviewed.

Chain of Custody Process



1. Data Collection

- First step in the process.
- Identify, label, record, and acquire data.
- Ensure the integrity of evidence during acquisition.

2. Examination

- Document chain-of-custody information.
- Record forensic procedures followed.
- Take screenshots to show steps completed and evidence found.

3. Analysis

- Uses legally acceptable methods to extract meaningful information.
- Answers case-related questions based on examination results.

4. Reporting

Includes:

- Statement about the chain of custody.
- Tools used in the investigation.
- Description of how data sources were analyzed.
- Identified issues and vulnerabilities.
- Suggested additional forensic steps.

Procedure for Establishing Chain of Custody : -

- Preserve original files.
- Photograph physical evidence.
- Take screenshots of digital evidence.
- Document date, time, and details of evidence receipt.
- Create a bit-for-bit clone of digital evidence on a forensic workstation.
- Perform hash verification on the clone to ensure accuracy.

Duplication and Preservation of Digital Evidence

Preserving the Digital Crime Scene

Secure the System First

- Before analyzing data, secure the compromised machine.
- Create a complete bit-stream backup of all storage devices.

Bit-Stream Backup

- More thorough than traditional backups.
- Copies every bit of data, including deleted files and slack space.
- For hard drives, create at least two bit-stream copies.
- All investigation work should be done only on the backup copy, not the original.

Tools Used for Bit-Stream Backups

IMDUMP

- First bit-stream backup tool created by Michael White.

SafeBack

- Widely used by intelligence, military, and law-enforcement agencies.
- Industry standard for forensic imaging.
- Captures all disk data, even hidden data in damaged clusters or sectors with bad CRCs.

SnapBack

- Another bit-stream imaging tool used by some law-enforcement teams.
- Very easy to use but much more expensive than SafeBack.
- Performs error-checking at every stage of backup and restore.

Computer Evidence Processing Steps

1. Turn the Computer Off

- Shut down or disconnect safely based on OS and network.
- Power off as quickly as possible when time is critical.

2. Document Hardware Setup

- Take photos of the system from all sides.
- Label all cables and connections for proper reassembly

3. Transport System to a Secure Location

- Never leave confiscated systems unattended.
- Store only in a secure, controlled environment.

4. Create Bit-Stream Backups

- Make forensic images of hard disks and floppy disks.
- Perform all analysis only on backup copies

5. Mathematically Verify Data

- Prove no evidence was altered after seizure.
- Use hash algorithms (used by law enforcement since 1989).

6. Record System Date & Time

- **Note incorrect clock settings or daylight-saving inconsistencies.**
- Important for accurate file timestamps.

7. Create a List of Keywords

- Collect keywords from experts related to the case.
- Use software to search disks using the keyword list.

8. Examine the Windows Swap File

- Swap file may contain useful data.
- File is deleted at shutdown → must be captured immediately.

9. Determine File Slack

- Contains leftover data/raw memory fragments.
- Important source of leads (often 80× more than swap file).

10. Examine Unallocated Space

- **Search erased file areas for relevant keywords.**

11. Keyword Search Across All Areas

- Search files, file slack, and unallocated space.
- Record all important findings from search results.

12. Document File Names, Dates & Times

- Record creation, modification, and access times.
- Create a properly formatted report for evidence logging.

13. Detect File/Program/Storage Irregularities

- Encrypted, compressed, or graphic files require manual review.
- Examine content for potential evidence.

14. Examine Software Operation

- **Run applications (if needed) to understand their function.**
- **Helps establish intent or malicious activity.**

15. Document All Findings

- Record every discovery, tool, procedure, and software version.
- Use screenshots to prove the exact process followed.

16. Retain Copies of Software Used

- Keep copies of tools/versions used during investigation.
- Ensures findings can be reproduced later if needed.

Legal Aspects of Collecting & Handling Digital Evidence

1. Proper Authorization

- Legal permission (e.g., warrant) is needed before accessing devices.
- Unauthorized collection makes evidence unusable.

2. Chain of Custody

- Record who handled the evidence, when, and how.
- Ensures evidence wasn't changed or tampered.

3. Compliance with Laws

- Follow laws like the IT Act or GDPR during collection and storage.
- Ensures evidence remains legally valid.

4. Preventing Contamination

- Use proper forensic tools and methods.

- Avoid changing or damaging the evidence.

5. Admissibility in Court

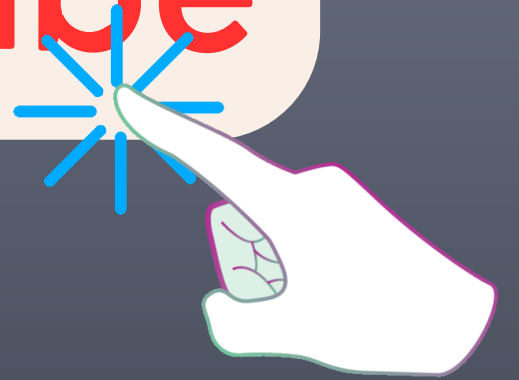
- Evidence must be legally obtained and relevant.
- Only properly handled evidence is accepted.

6. Secure Storage

- Store evidence safely and control access.
- Prevents unauthorized use or manipulation.

7. Expert Handling

- Only trained forensic experts should collect and analyze evidence.
- They may need to explain it in court.



THANK
YOU